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Evidence-based rehabilitation following anterior cruciate ligament reconstruction

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Abstract Following a bone-patellar tendon-bone autograft (BPTB) or four-stranded semitendinosus/gracilis tendons autograft (ST/G) anterior cruciate ligament (ACL) reconstruction, the speed and safety with which an athlete returns to sports (or regains the pre-injury level of function) depends on the rehabilitation protocol. Considering the large differences in clinical and outpatient protocols, there is no consensus regarding the content of such a rehabilitation program. Therefore, we conducted a systematic review to develop an optimal evidence-based rehabilitation protocol to enable unambiguous, practical and useful treatment after ACL reconstruction. The systematic literature search identified 1,096 citations published between January 1995 and December 2006. Thirty-two soundly based rehabilitation programs, randomized clinical trials (RCT's) and reviews were included in which common physical therapy modalities (instruction, bracing, cryotherapy, joint mobility training, muscle-strength training, gait re-education, training of neuromuscular function/balance and proprioception) or rehabilitation programs were evaluated following ACL reconstruction with a BPTB or ST/G graft. Two reviews were excluded because of poor quality. Finally, the extracted data were combined with information from background literature to develop an optimal evidence-based rehabilitation protocol. The results clearly indicated that an accelerated protocol without postoperative bracing, in which reduction of pain, swelling and inflammation, regaining range of motion, strength and neuromuscular control are the most important aims, has important advantages and does not lead to stability problems. Preclinical sessions, clear starting times and control of the rehabilitation aims with objective and subjective tests facilitate an uncomplicated rehabilitation course. Consensus about this evidence-based accelerated protocol will not only enhance the speed and safety with which an athlete returns to sports, but a standardized method of outcome measurement and reporting will also increase the evidential value of future articles.

Keywords Bone-patellar tendon-bone graft · Tendon transfer · Physiotherapy · Accelerated protocol · IKDC questionnaire · Systematic review

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Introduction

After back complaints, knee injuries are the most frequent problems of the musculoskeletal system reported in primary care. The prevalence is 48 per 1,000 patients per year [35]. In 9% of these cases, there is damage to one or more ligaments, of which the anterior cruciate ligament (ACL) is the most commonly injured. Most ACL ruptures occur during sports activities in the age group of 15-to-25-yearold athletes. The injury mechanism is a valgus/external rotation trauma with a slightly bend knee [3, 18, 35, 49].

Because the ACL is a primary stabilizer of the knee, a rupture can lead to functional instability (i.e., giving-way episodes). Conservative or surgical treatment of this instability is indicated for regaining pre-injury level of function [3, 4, 18, 35, 42, 49]. In the long term, an ACL rupture can cause further intraarticular damage like meniscal tears, cartilage defects and osteoarthritis. Two-thirds of primarily conservative treated patients opt for an ACL reconstruction after rehabilitation [47]. The younger and more active the patient, the earlier surgical reconstruction is chosen [3, 4, 10, 18, 31, 35, 42, 48].

A recent meta-analysis showed that there were only marginal clinical and functional differences in outcome between the two most commonly used surgical techniques, bone-patellar tendon-bone autograft (BPTB) and four-stranded semitendinosus/gracilis tendons autograft (ST/G). Due to improved fixation methods, despite the longer incorporation time of the ST/G method (12 instead of 8 weeks) [4, 6, 18, 20, 24, 51], no significant differences in stability (elongation or rupture of the graft) were found [12].

After ACL reconstruction, the speed and safety with which an athlete returns to sports or regains the pre-injury level of function depends largely on the rehabilitation protocol [5, 6, 49]. There is no current consensus regarding the content of such a rehabilitation program, hence we conducted a systematic review. The purpose of this review was to design an optimal evidence-based accelerated rehabilitation protocol (return to sport within 6 months, see Appendix 1) following ACL reconstruction. While no evidence for elongation or rupture of the BPTB or ST/G graft during accelerated rehabilitation exists, important advantages are described compared to more conservative programs (return to sports in 9–12 months): lower costs, earlier return to sports and graft healing, earlier recovery of range of motion (ROM), knee function and muscle strength, fewer complications like arthrofibrosis [2, 4, 11, 12, 23, 24, 34, 40, 43, 51].

Materials and methods

A computerized literature search (Table 1) was performed using the Cochrane Library, MEDLINE (PubMed), EMBASE and PEDro to identify relevant articles published between January 1995 and December 2006. Based on information from the title, abstract and full text, citations were considered for inclusion if they met the inclusion and exclusion criteria. Reference lists of the included studies were reviewed for additional publications.

Inclusion criteria

- Soundly based rehabilitation programs (protocols based on an extensive search of the literature), RCT's and reviews testing interventions or exercise programs designed to rehabilitate ACL reconstruction.
- ACL reconstruction with a BPTB or ST/G graft.

Included physical therapy interventions

- instruction (surgery, complications, rehabilitation program, exercises)
- bracing
- cryotherapy
- joint mobility training (active, active assisted and resisted)
- muscle-strength training (isometric, isotonic, isokinetic; concentric, eccentric; open kinetic chain, closed kinetic chain)
- gait re-education
- training of neuromuscular function/balance and proprioception

Excluded physical therapy interventions

- hydrotherapy
- electrotherapy (i.e., ultrasound, transcutaneous electrical nerve stimulation (TENS), muscle stimulation)
- complimentary therapies such as reflexology

To assess the methodological quality of the RCT's and reviews, checklists of the Cochrane Library were used. The checklists are available at http://www.cochrane.nl. Every article considered for inclusion was given a final judgment (good, questionable, poor) regarding estimated validity and clinical relevance (applicability). Publications of poor quality were excluded. Next, per included article data were systematically extracted regarding research question, included patients, surgical technique used, physical therapy intervention, outcome measures, results, author's conclusion and/or final rehabilitation protocol.

The search strategy, selection of articles, quality assessment and data extraction were conducted independently by two reviewers (SvG and CH). Disagreements were resolved by consensus. If consensus was not reached the final decision was made by a third reviewer (CvL). Finally, the information from background literature, soundly based rehabilitation programs, RCT's and reviews was combined by the two reviewers (SvG and CH) to develop an optimal and evidence-based rehabilitation protocol following ACL reconstruction.

Table 1 Search strategy

	Citations medline	Citations embase	Citations cochrane	Citations pedro	Total citations	Total exclusion	Total inclusion
1: Anterior cruciate ligament	9468						
2: ACL	6209						
3: 1 or 2	11599	6094	600	98			
4: Reconstruction	121234						
5: Ligament surgery	25921						
6: Tendon graft	3303						
7: Interference screw	650						
8: Orthopedic procedures	153078						
9: Tendon transfer	3647						
10: 4 or 5 or 6 or 7 or 8 or 9	274181	72112	2798				
11: Physiotherapy	104030						
12: Physical therapy	165739						
13: Postoperative care	90306						
14: Rehabilitation	269203						
15: Intervention	218508						
16: Instruction	144080						
17: Exercise movement techniques	3932						
18: Exercise therapy	54357						
19: Exercise	180416						
20: Kinesiotherapy	66						
21: 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20	933984	156693	8049				
22: 3 and 10 and 21	1916	1004		98			
23: 22 and Limits[a]	1068	701	37				
Total citations 4 databases	(1068)	(701)	(37)	(98)	1904		
Total relevant citations based on title[b]					*1096*		
Excluded based on abstract[b]						1024	
Excluded based on full text[b]						18	
Excluded due to poor quality						2	
Included citations							32
Added background information (articles)							18
Added background information (book and online document)							2

[a] English, French, German, Dutch, Humans, published between 1995–2006

[b] Considering: article type, graft type, physical therapy modality/rehabilitation program

Results

The structured search strategy identified 1,096 relevant citations. Based on the inclusion criteria, 32 rehabilitation programs, RCT's or reviews were finally included (English: 30, German: 2) and 20 articles with background information were added (English: 17, German: 1, Dutch: 2). Only two reviews were excluded due to poor quality. Any disagreement between the two reviewers (SvG and CH) was resolved by consensus.

Presurgery

There is consensus in the literature about the optimal timing of surgery. To prevent delay in recovery caused by postoperative complications like arthrofibrosis, surgery should not be initiated before achievement of certain preoperative goals: minimal pain, swelling and inflammation response, full ROM and neuromuscular control (i.e., of optimization muscle strength and gait pattern) of the injured extremity [1, 3, 8, 18, 24, 35, 46, 51]. This is why the Dutch physician standard advocates that ACL reconstruction should not be timed within 6–8 weeks after trauma [35].

Clear instruction (postsurgical exercises, walking with crutches) and information about the content of the rehabilitation program increases self-efficacy during rehabilitation, stimulates early recovery of knee function, decreases expected postsurgical pain and creates a realistic view about the rehabilitation process [6, 8, 22, 51].

Postsurgery, phase 1 (week 1)

In order to prevent postsurgical complications, the most important goals in phase 1 are controlling pain, swelling and inflammation, recovery of ROM and neuromuscular control [6, 8, 46, 51]. There are no long-term advantages of bracing [4, 13, 16, 26, 28, 33, 39, 52]. Aggressive control of pain, swelling and inflammation prevents quadriceps inhibition, maintains full knee extension and makes immediate weight bearing possible [1, 6, 24, 51]. In addition to medication, exercises, postsurgical compression wraps and elevation, cryotherapy is advised as it significantly reduces postsurgical pain [37].

Immediate recovery of passive and active ROM (with an emphasis on full extension) following ACL reconstruction reduces pain, stimulates the homeostasis of cartilage and prevents patellofemoral problems, alterations in gait pattern, quadriceps atrophy and arthrofibrosis. Multidirectional mobilizations of the patella should be included because patellar immobility leads to decreased ROM and quadriceps inhibition [1, 4, 6, 8, 11, 24, 36, 46, 51].

Without endangering the ACL graft, muscle control can be regained and should be initiated in phase 1 by isometric, closed chain (CC, safe range 0°–60°) and open chain (OC, safe range 90°–40°) exercises without additional weight. These exercises should include muscle setting exercises (MSE), straight leg-raising (SLR), heel slides, mini squads (0°–30° flexion), shifting body weight, OC extension (90°–40°) and OC flexion (isolated hamstring) exercises [6, 8, 11, 24, 36, 40, 42, 45, 51].

Full weight-bearing without crutches within 10 days (with a normal gait pattern) improves quadriceps function, prevents patellofemoral pain and does not affect knee stability. Pain, swelling, insufficient ROM and quadriceps weakness are the most common causes for an altered gait pattern [4, 6, 11, 36, 40, 51].

Phase 2 (week 2 to week 9)

Cryotherapy should be continued because persistence in pain, swelling and inflammation may result in postsurgical complications like decreased ROM, decreased quadriceps control, altered gait pattern and a prolonged rehabilitation process [6, 24, 41, 51]. Flexion can be increased gradually while full extension and patellar mobility will be maintained. Inadequate progression of ROM extension can be treated aggressively to prevent postsurgical complications like arthrofibrosis [6, 8, 24, 36, 43, 51].

In phase 2, the strength of the graft is not optimal [4, 6, 18, 20, 24, 51]. Quadriceps and hamstring strength can be increased by isometric isotonic and isokinetic exercises without endangering the graft. Because of the advantages mentioned in the literature, isokinetic exercises are advised if adequate equipment is available [4, 6, 8, 11, 24, 36, 40, 42, 45, 50, 51]. Isotonic strength training in a safe range (CC: 0°–60°, OC: 90°–40°), aimed at endurance, increases quadriceps strength significantly and has no negative effect on anterior knee pain and knee laxity [4, 24, 42, 49, 51]. There is, both for BPTB as for ST/G reconstructions, increasing evidence of the safety of CC and OC exercises exceeding the safe range (CC: 0–90°, OC: 90°–0°) [9, 17, 27, 29, 30, 32]. Quadriceps atrophy, persistent quad lag with SLR, incomplete extension and gait impairments in week 5 are predisposing factors for quadriceps weakness after 6 months [36].

Despite limited evidence with variation in results, it is generally accepted that loss of proprioception in ACL-deficient knees occurs and that neuromuscular training is essential for functional recovery following ACL reconstruction and secondary prevention (re-rupture) [4, 6, 7, 21, 24, 38, 40, 41, 43, 49, 51]. Neuromuscular training should start as soon as walking without crutches is possible, with gentle non-complex exercises using minimal weight and developing from static to dynamic balance training and plyometric exercises into agility training and sport-specific exercises [4, 6, 7, 21, 24, 38, 40, 41, 43, 49, 51].

Gait training on a treadmill or flat surface without crutches is still necessary in phase 2, because a protective gait pattern can still exist despite a normal gait pattern at first sight [6, 36]. Specific exercises for phase 2 should include walking on a treadmill, cycling on an ergometer and swimming from week 3, stair-stepping machine from week 4, jogging in a straight line and outdoor cycling from week 8 [6, 24, 25, 40, 41, 43, 51].

Phase 3 (week 9 to week 16)

To prevent postoperative complications like arthrofibrosis, obtaining and maintaining full ROM remains an important goal [6, 8, 24, 36, 43, 51]. Because the tensile strength of the graft is rising in this phase, muscle strength of the knee stabilizers can be increased further with CC and OC exercises [4, 6, 18, 20, 24, 51]. Pain and swelling determine the progression from endurance (many repetitions/no additional weight) to more resistance training (fewer repetitions/increasing weight). CC and OC exercises form the ideal basis for sport-specific functional training in phase 4 [32, 36, 42, 51].

Neuromuscular control can be further improved by slowly increasing functional dynamic balance training and plyometric exercises. Training of functional movement patterns improves the interaction between stabilizing structures of the kinetic chain (trunk, hip, knee and ankle) [51]. Plyometric exercises are an appropriate preparation for agility training (phase 4) because they improve the concentric contraction power of the muscle so that quicker changes in direction are possible [6, 41]. To stimulate coordination and control through afferent and efferent information processing, exercises should be enhanced by variation in visible input, surface stability, speed of exercise performance, complexity of the task, resistance, one- or two-legged performance, etc. [6, 38, 41, 51]. Specific exercises for phase 3 should include normalization of running (gradually increasing duration and speed to decrease neuromuscular adaptation and recovery time) from week 9, jogging outdoors starts in week 13 [6, 24, 41].

Phase 4 (week 16 to week 22)

Maximizing endurance and strength of the knee stabilizers, optimizing neuromuscular control with plyometric exercises, agility training and sport-specific exercises are the essential goals of this phase. Sport-specific agility training with variations in running, turning and cutting maneuvers, acceleration and deceleration, improves arthrokinetic reflexes so that new trauma during competition can be prevented [6, 38, 41, 51].

Intermediate-phase tests and return to sports criteria

To evaluate periodical recovery following ACL reconstruction (pain, swelling, ROM, strength and neuromuscular control) and to correctly time return to sports, the literature advises using tests that are reliable, valid, responsive to change with time, feasible and clinically relevant [6, 44]. The following tests fulfill these qualities and should therefore be used in an evidence-based rehabilitation protocol (for timing and performance see Appendix 2):

- Visual Analog Scale (VAS): for pain [44].
- Circumference measurement with a measuring tape: despite lack of validating studies, it appears to measure swelling following ACL reconstruction. The non-involved leg is used as a control for measurement standardization per person [44].
- Goniometer: for notation of active and passive ROM measurements [44].
- The International Knee Documentation Committee Subjective Knee Form (IKDC): a knee-specific questionnaire designed to measure symptoms and limitations in function after knee-related ligament trauma [14].
- Hop tests: measures total leg function. To detect performance limitations, a minimum of two hop tests should be performed to increase sensitivity. The test just measures differences between the involved and uninvolved leg. It is not a diagnostic tool for possible underlying causes (lack of strength, confidence or neuromuscular control) [44].
- Isokinetic tests: objective measurement of strength and endurance of the knee stabilizers requires proper description of variables relating to subjects (e.g., age, gender, weight) and test procedure (e.g., range, type of muscle contraction, pre-test procedures, test conditions and type of data analysis), because they significantly affect the result of the measurement [19, 44].

If the rehabilitation goals of the previous phase are met, the next phase can be started. Patients can return to sports if full ROM is achieved, the hop tests and strength of the hamstrings and quadriceps are at least 85% compared to the contralateral side, the difference in hamstring/quadriceps strength ratio is less than 15% compared to the contralateral side, and when the patient tolerates sport-specific activities (no increase in pain and swelling). Hop tests and isokinetic tests can only be performed if the knee is stable in active situations [6, 11, 24, 41, 44, 51].

See Table 2 for an overview of the results of the included RCT's and reviews on specific topics (accelerated versus conservative rehabilitation programs, BPTB versus ST/G, bracing, cryotherapy, immediate versus delayed motion, immediate versus delayed weight-bearing, early strength restoration, closed chain versus open chain exercises, neuromuscular training).

Discussion

The most important finding of the presented study was that, although many authors published about ACL rehabilitation, there is an urgent need for consensus about an up to date, gapless and as evidence-based as possible protocol to enhance the speed and safety in which an athlete returns to sport and to increase the evidential value of future articles.

The presented accelerated rehabilitation protocol following BPTB or ST/G ACL reconstruction (with boundaries, clear starting times and control of the rehabilitation aims with objective and subjective tests to facilitate an uncomplicated rehabilitation course) is based on scientific evidence regarding complications such as graft elongation or rupture and the advantages mentioned in the literature. An optimal rehabilitation program includes preoperative patient education and training to create a realistic view, enhance independence and facilitate an optimal timing of surgery [1–4, 6, 8, 11, 12, 22–24, 34, 35, 40, 43, 44, 46, 51].

The effect of an accelerated rehabilitation protocol with regard to the most used fixation methods (interference screws and endobutton) has been documented extensively [12, 34]. Although there is no significant difference in

Table 2 Results specific topics of inlcuded RCT's and reviews

Author (language)	Year	Study design	Materials & methods	Results	Method. quality
				Accelerated versus conservative rehabilitation programs	
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	Recent studies of healing BPTB grafts indicate that accelerated rehabilitation (19 weeks) produces the same clinical, functional and patient-oriented outcomes compared to non-accelerated rehabilitation (32 weeks)	+
Risberg et al. (E)	2004	Review	Review of the effectiveness of various rehabilitation programs in RCT's used for surgically or non-surgically treated ACL injuries	Aggressive rehabilitation appears to result in better graft healing than reported in animal studies. There is clinical evidence that laxity does not increase following aggressive rehabilitation programs. There are no significant differences at 12 months between a 6- and 8-month revalidation program following ACL reconstruction, except for the earlier return to sport in the 6-month group (6 vs. 9 months)	+
Shaw (E)	2002	Review	Review (RCT's or best evidence) of the evidence for the efficacy and safety of accelerated rehabilitation programs (4–6 months) following ACL reconstruction (BPTB or ST/G)	There is increasing evidence indication that accelerated rehabilitation following ACL reconstruction is effective in restoring knee ROM and muscle strength and returning the athlete to sporting activity more expediently than traditional programs. There is little compelling evidence that accelerated rehabilitation is unsafe, with regard to stability or compromise of the graft	±
Beynon et al. (E) Reference [2]	2005	RCT	Accelerated rehabilitation 19 weeks versus non-accelerated rehabilitation 32 weeks following ACL reconstruction (BPTB). Two-year follow-up. ($N = 25$)	Both programs increase anterior knee laxity and have the same effect in terms of clinical assessment, patient satisfaction, functional performance	+
				BPTB versus ST/G	
Goldblatt et al. (E)	2005	Meta-analysis	Meta-analysis of 11 studies (randomized or sequential inclusion) comparing the effectiveness of ACL reconstruction using either BPTB or ST/G) grafts. Minimum of 2-year follow-up	The incidence of instability is not significantly different between BPTB and ST/G grafts	+
				Bracing	
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	Extended immobilization of the knee after ACL reconstruction is detrimental to the structures that surround the knee (ligaments, cartilage, bone and musculature). At longer follow-up bracing or functional bracing has no effect on outcome (ROM, subjective outcome, knee laxity, activity level or function)	+
Harilainen et al. (E)	2006	RCT	Bracing 12 weeks versus no bracing following ACL reconstruction. 80% 5-year follow-up. ($N = 60$)	There were no differences between the groups (knee-scores, activity level, degree of laxity, isokinetic peak muscle torque)	±
Pförringer et al. (G)	2005	RCT	Bracing in full extension 3 days versus bracing in 20° flexion 3 days, following ACL reconstruction. Follow-up of 1 year. ($N = 46$)	Immediate postoperative extension, compared to postoperative flexion of 20°, reduces the term of rehabilitation and optimizes the operative results after ACL reconstruction	+
Mikkelsen et al. (E)	2003	RCT	Bracing in hyperextension (−5°) versus bracing in full extension (0°) for 3 weeks, following ACL reconstruction. Follow-up of 3 months. ($N = 44$)	No significant differences were found between the groups in terms of flexion, laxity and postoperative pain, but there was significant less loss of extension in the hyperextension bracing group after 3 months. Postoperative extension can by reached by other means	±

Table 2 continued

Author (language)	Year	Study design	Materials & methods	Results	Method. quality
Hendriksson et al. (E)	2001	RCT	Range of motion training in brace (starting day 7) versus immobilization for 5 weeks, following ACL reconstruction. Follow-up of 2 years. ($N = 50$)	Postoperative bracing treatment with early range of motion training after ACL reconstruction gave as good ROM, knee stability, subjective knee function and activity level as immobilization	±
Möller et al. (E)	2001	RCT	Bracing 6 weeks versus no bracing following ACL reconstruction. 90% follow-up of 2 years. ($N = 62$)	No differences were found between the groups in either subjective or objective knee stability of patients knee function in any stage up to 24 months after surgery	+
Wu et al. (E)	2001	RCT	Ability to reproduce knee joint angles and isokinetic performance at 5 months after ACL reconstruction. Bracing, versus placebo bracing, versus no bracing. ($N = 31$)	Bracing can improve static proprioception of the knee joint, but not the muscle contractile function, under isokinetic testing conditions. The bracing and placebo bracing group had similar performances for joint angle reproduction (proprioception), so it is not caused by mechanical restriction	±
Risberg et al. (E)	1999	RCT	Bracing (2 weeks rehabilitative brace and 10 weeks functional brace) versus no bracing following ACL reconstruction. Follow-up of 2 years. ($N = 60$)	There were no significant differences with regard to knee joint laxity, ROM, muscle strength, functional knee tests, patient satisfaction or pain	+
				Cryotherapy	
Raynor et al. (E)	2005	Meta-analysis	Meta-analysis combining the scientific evidence of 7 RCT's evaluating the effectiveness of cryotherapy compared to a placebo group, following ACL reconstruction	Cryotherapy has a statistically significant benefit in postoperative pain control. As it is fairly inexpensive, easy to use, has a high level of patient satisfaction and has rarely adverse events, cryotherapy is justified in the postoperative management of ACL reconstruction	+
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	Only 1 study was identified (cooling pad, crushed ice or no cryotherapy): cooling pad and crushed ice treatments were found to produce a significant decrease in knee temperature. No differences regarding duration of hospital stay, ROM or use of pain medication were found	+
				Immediate versus delayed motion	
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	There is little doubt that early joint motion after ACL reconstruction is beneficial: it leads to pain reduction, lessens adverse changes in articular cartilage and helps prevent the formation of scar and capsular contractions that have the potential to limit joint motion	+
Shaw (E)	2002	Review	Review (RCT's or best evidence) of the evidence for the efficacy and safety of accelerated rehabilitation programs (4–6 months) following ACL reconstruction (BPTB or ST/G)	One of the primary objectives of ACL rehabilitation is to restore functional ROM. Accelerated programs, with early ROM restoration, achieve significantly earlier and greater restoration of flexion and extension ROM than patients in a traditional rehabilitation program	±
				Immediate versus delayed weight-bearing	
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	Immediate weight-bearing after ACL reconstruction produces similar clinical patient and functional outcomes to delayed weight-bearing and no increase in anterior knee laxity. Immediate weight-bearing may be beneficial because it lowers the incidence of anterior knee pain	+

Table 2 continued

Author (language)	Year	Study design	Materials & methods	Results	Method. quality
Risberg et al. (E)	2004	Review	Review of the effectiveness of various rehabilitation programs in RCT's used for surgically or non-surgically treated ACL injuries	Immediate weight-bearing after ACL reconstruction can be recommended because it does not compromise knee joint stability and results in diminished anterior knee pain	+
Shaw (E)	2002	Review	Review (RCT's or best evidence) of the evidence for the efficacy and safety of accelerated rehabilitation programs (4–6 months) following ACL reconstruction (BPTB or ST/G)	Early postoperative weight-bearing advantages are cartilage nutrition, facilitation of collagen reorganization during healing and allowing osseous and soft tissue of the knee to respond to normal physiological loading	±
				Early strength restoration	
Shaw et al. (E)	2005	RCT	Early quadriceps exercises group versus no quadriceps exercises group for first 2 weeks, following ACL reconstruction. Follow-up of 6 months. ($N = 103$)	Isometric quadriceps exercises and straight leg raises can be safely prescribed during the first two postoperative weeks and confer advantages for faster recovery of knee ROM and stability	+
				Closed chain versus open chain exercises	
Trees et al. (E)	2006	Review	Review (RCT's and quasi-RCT's) evaluating the effectiveness of exercises following ACL reconstruction	No differences between closed and open chain exercises in knee function of knee laxity measured were found at 1 year. At 31 months after surgery, return to pre-injury level of sport was statistically more common in a closed and open chain exercise group compared to closed chain exercises alone. There were no differences in knee laxity and isokinetic quadriceps strength at 6 months postsurgery. However, due to methodological and reporting differences in the trials it is not possible to support the efficacy of one exercise intervention over the other	+
Beynon et al. (E) Reference [4]	2005	Review	Review (RCT's or best evidence) of technical aspects of ACL surgery, bone tunnel widening, graft healing and rehabilitation after ACL reconstruction	Different open and closed kinetic chain rehabilitation programs were compared over different times, and it is therefore impossible to come to a consensus regarding the effectiveness of one approach compared to another	+
Fleming et al. (E)	2005	Review	Review (RCT's and biomechanical studies) of the effects of closed chain and open chain exercises on graft healing, following ACL reconstruction	Open and closed chain exercises may not differ in their effects on the healing response of the ACL reconstructed knee. Recent biomechanical studies have shown that the peak strain produced on a graft is similar. Clinical studies suggest that both play a beneficial role in the early rehabilitation of the reconstructed knee	±
Risberg et al. (E)	2004	Review	Review of the effectiveness of various rehabilitation programs in RCT's used for surgically or non-surgically treated ACL injuries	The concerns about possible negative effects on anterior knee pain and joint laxity while using open chain exercises for quadriceps strengthening were not supported by the results from RCT's. These data indicate that both produce the same amount of strain on the ACL and can be introduced as soon after reconstruction as weight-bearing exercises. Open chain exercises seem to be favorable for increasing quadriceps strength. In order to minimize strain on the graft, the knee should be maintained in less than 60° during closed chain exercises and open chain exercises with knee angles greater than 40° of flexion are recommended	+

Table 2 continued

Author (language)	Year	Study design	Materials & methods	Results	Method. quality
Perry et al. (E)	2005	RCT	Closed chain versus open chain exercises, for quadriceps strengthening, starting at week 8–14 after ACL reconstruction. Follow-up at week 14. ($N = 49$)	Closed and open chain quadriceps training in the middle period of rehabilitation after ACL reconstruction do not differ in their effects on knee laxity or leg function	+
Morrissey et al. (E)	2002	RCT	Closed chain versus open chain exercises, for quadriceps strengthening, starting at 2–6 weeks after ACL reconstruction. Follow-up at week 6. ($N = 43$)	Open chain and closed chain exercises in the early period after ACL reconstruction do not differ in their immediate effects on anterior knee pain	±
Hooper et al. (E)	2001	RCT	Closed chain versus open chain exercises, for quadriceps strengthening, starting at week 2–6 after ACL reconstruction. Follow-up at week 6. ($N = 37$)	There are no clinically significant differences in the functional improvement resulting from the choice of open or closed chain exercises in the early period after ACL reconstruction	+
Mikkelsen et al. (E)	2000	RCT	Closed chain exercised versus combined open and closed chain exercised for quadriceps strengthening starting at week 6 after ACL reconstruction. Follow-up of 6 months. ($N = 44$)	The addition of open quadriceps training after ACL reconstruction results in a significantly better improvement in quadriceps torque without reducing knee joint stability at 6 months and also leads to a significantly higher number of athletes returning to their previous activity earlier and at the same level as before injury	±
Morrissey et al. (E)	2000	RCT	Closed chain versus open chain exercises, for quadriceps strengthening, starting at week 2–6 after ACL reconstruction. Follow-up at week 6. ($N = 36$)	There were no significant differences between the groups at 6 weeks. The great concern about the safety of open chain exercises may not be well founded	+
				Neuromuscular training	
Risberg et al. (E)	2004	Review	Review of the effectiveness of various rehabilitation programs in RCT's used for surgically or non-surgically treated ACL injuries	The evidence for neuromuscular training is limited, but all trials have reported promising results both for increased knee function and increased neuromuscular control. Although we lack dose–response evidence and long-term follow-up studies, clinicians should incorporate neuromuscular training	+
Cooper et al. (E)	2005	RCT	Proprioceptive and balance exercise program versus a strengthening program first 6 weeks after ACL reconstruction. Follow-up at 6 weeks. ($N = 29$)	There were no significant differences between groups on hop testing at 6 weeks. For several items in the Cincinnati knee rating system and the patient-specific functional scale, the strengthening group improved more. There is either no difference between the groups or a strengthening program is more beneficial in the early phase of rehabilitation	+
Liu-ambrose et al. (E)	2003	RCT	Proprioceptive training program versus isotonic strength training program for 12 weeks, after ACL reconstruction. Follow-up at 12 weeks. ($N = 10$)	The proprioceptive group demonstrated greater percent change in isokinetic torques. Both groups demonstrated similar significant gains in functional ability and subjective scores. Proprioceptive training alone can induce isokinetic strength gains	

Language: English (E), German (G). Final judgement methodological quality of included RCT or review is good (+), questionable (±) or poor (−)

stability between a BPTB and ST/G graft, after an accelerated protocol, subtle differences in laxity, kneeling pain, ROM and loss of extension/flexion have been reported [12]. Different fixation methods could also lead to other clinical and functional outcomes. Good additional RCT's with an accelerated rehabilitation program following BPTB and ST/G reconstructions are necessary to determine long-term results with other fixation methods.

Close collaboration between orthopedic surgeon, physical therapist and patient is necessary for successful recovery following ACL reconstruction. Accurate monitoring of progress makes early intervention possible and prevents postoperative complications like arthrofibrosis. Patients can start with the next phase only if the rehabilitation goals of the previous phase are met and confirmed by the objective and subjective tests. The suggested time frame of 22 weeks may need to be customized per patient [6, 8, 20, 24, 40, 44, 45, 51].

Despite the fact that the protocol presented in this review is based on information from background literature, soundly based rehabilitation protocols, RCT's and reviews from the four most important databases, the evidence is not conclusive. Considering the lack of in vitro studies, uncertainties about the recovery mechanism of the graft following ACL reconstruction will remain [4, 6, 18, 20, 24, 51]. Sometimes additional evidence is not yet available. Despite increasing consensus that OC exercises (in and exceeding the safe range) with a focus on endurance do not increase graft laxity and have favorable effects on quadriceps strength, there is, especially when using a ST/G graft, still uncertainty about the optimal timing of introduction of these OC exercises [4, 9, 15, 17, 24, 27, 29, 30, 32, 42, 49, 51]. Additional studies are necessary to determine which combination of CC and OC exercises optimize quadriceps strength most efficiently and to better estimate the consequences of early introduction of OC exercises with ST/G grafts.

We realize that we added evidence, with lower level and methodological quality, from soundly based protocols and available background literature (descriptive reviews and articles, lectures, clinical controlled trials, book and online Dutch physician guideline) to the high level evidence from our systematic review (RCT's and reviews). Without the added information (consensus assessment if there were no RCT's or reviews available, background information about graft healing, avoiding arthrofibrosis and other complications, valid and reliable tests, etc.), we would not have been able to develop a continuous rehabilitation protocol, leaving the orthopedic surgeon and physiotherapist with gaps in scientific evidence after BPTB or ST/G ACL reconstruction.

This strategy is fully in line with the findings by Beynnon et al. [4]. Based on their extensive research, they stated that there is little consensus in the literature about what composes an accelerated rehabilitation program, because very few of the identified RCT's, described the rehabilitation protocol adequately. Trees et al. [49] could not support one form of intervention against another because of the absence of a standardized method of outcome measurement and reporting. They advocated for international consensus to increase the evidential value of future trials.

Conclusion

The presented results clearly indicate that an accelerated protocol without postoperative bracing, in which reduction of pain, swelling and inflammation, regaining range of motion (ROM), strength and neuromuscular control are the most important aims, has important advantages and does not lead to stability problems. Preclinical sessions, clear starting times and control of the rehabilitation aims with objective and subjective tests facilitate an uncomplicated rehabilitation course. Consensus about this evidence-based accelerated protocol will not only enhance the speed and safety with which an athlete returns to sports, but a standardized method of outcome measurement and reporting will also increase the evidential value of future articles.

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Appendix 1: Rehabilitation protocol following ACL reconstruction (BPTB and ST/G method)

Presurgery

Preferably patients will be seen by the physical therapist at least three times presurgically. In these sessions, the following will be considered:

- Information about the rehabilitation (discuss mutual expectations). Emphasize that knee rehabilitation is more than strength-training of the upper-leg muscles alone. The (neuromuscular) rehabilitation addresses the whole lower extremity, core stability and training of the kinetic chain.
- Decrease of pain, swelling and inflammation.
- Achieve/maintain normal range of motion (ROM) with a focus on good patellar mobility.
- Achieve/maintain normal gait pattern.
- Maintain muscle strength, prevent atrophy.
- Training of the first-days postoperative exercises (i.e., heel slides, straight-leg raising (SLR), squads, leg elevation with a pillow under the heel). Emphasize the importance of full extension.
- Practice alternate walking with crutches for the first days postoperatively.
- Tests: VAS-score pain; ROM; measurement of knee swelling; IKDC questionnaire; hop tests and optional isokinetic tests for strength and endurance of the knee flexors and extensors.

Phase 1 (week 1)

- Control of pain and inflammation (i.e., through cryotherapy and exercises).
- Obtain ROM of 0°–90°, emphasizing achievement of full extension (i.e., through CPM and exercises: patellar

mobilization in all directions, heel slides and leg elevation with a pillow under the heel).

- Regain muscle control, with safe isometric and isotonic OC (ROM 90°–40°) and CC (ROM 0°–60°) strength exercises without additional weight. (i.e., SLR, mini squads, shifting body weight).
- Improve gait pattern. If pain is tolerated, aim at walking without crutches from day 4. Sufficient neuromuscular control and a non-limping gait pattern are criteria for walking without crutches.

Criteria to start with phase 2

- Pain knee is equal to previous week or less (VAS-score pain).
- Minimal swelling (measurement with measuring tape).
- Full extension and 90° flexion are possible (ROM-goniometer).
- Good patellar mobility compared with contralateral side.
- Sufficient quadriceps control to perform a mini squad 0°–30° and SLR in multiple directions.
- Ability to walk independently with or without crutches.

Phase 2 (week 2 to week 9)

- Apply cryotherapy in case of pain or swelling (if necessary after each therapy session).
- Work toward full ROM (maintain full extension, 120° flexion from week 2 and 130° flexion from week 5) with remaining attention for good patellar mobility.
- Walking without crutches from day 4 to 10. Normalize gait pattern with walking exercises (treadmill from week 3 and jogging in a straight line from week 8).
- Isometric and isotonic strength training increasing in intensity (quadriceps, hamstring, gastrocnemius and soleus), with increasing ROM for OC and CC exercises without extra weight. For OC exercises: weeks 2, 3 and 4 from 90° to 40°, afterward 10° toward extension to be added every week. For CC exercises: weeks 2–7 from 0° to 60° and from week 8 from 0° to 90°.
- Start neuromuscular training by slowly increasing from static stability to dynamic stability. Work toward confidence on the vestibular and somatosensory system for balance, with increasing surface instability and decreasing visual input.
- Start from week 3 with cycling on an ergometer and swimming.
- Start from week 4 with stepping on a stair-stepping machine.
- Start from week 8 with outdoor cycling.

Caution: act adequately in case of persisting pain, inflammation or limited ROM → there is a risk of developing arthrofibrosis (in case of doubt consult the orthopedic surgeon)!

Criteria to start with phase 3

- Minimal pain and swelling (VAS-score pain, measurement of knee swelling with measuring tape).
- Full extension and at least 130° flexion possible (ROM-goniometer).
- Normal gait pattern.
- Exercises of previous week are carried out properly.
- Administer the IKDC questionnaire.

Phase 3 (week 9 to week 16)

- Obtaining and maintaining full ROM.
- Optimizing muscle strength and endurance. Add increasing weights from week 9 both for OC and CC exercises.
- Neuromuscular training with increasing emphasis on dynamic stability and plyometric exercises, slowly increasing duration and speed. Start with two-legged jumping and work slowly toward one-legged jumping. Normalize running with outdoor jogging from week 13.

Criteria to start with phase 4

- No pain or swelling in the knee (VAS-score pain, measuring knee swelling with measuring tape).
- Full flexion and extension of the knee (ROM-goniometer).
- Administer the IKDC questionnaire again.
- Quadriceps and hamstring strength >75% compared to the contralateral side. Difference in hamstring/quadriceps strength ratio is ≤15% compared to the contralateral side (optional isokinetic strength testing of knee flexors and extensors at 180° per second).
- Hop tests >75% compared to the contralateral side.
- Exercises of previous week are carried out properly.

Phase 4 (week 16 to week 22)

- Maximizing muscle endurance and strength.
- Maximizing neuromuscular control with emphasis on jumping, agility training and sport-specific tasks. Variations in running, turning and cutting maneuvers are allowed. Duration and speed to be increased and maximized.

Criteria for returning to sports

- No pain or swelling (VAS-score pain, measuring knee swelling with measurement tape).
- Full flexion and extension of the knee is possible (ROM-goniometer).
- Quadriceps and hamstring strength >85% compared to the contralateral side. Difference in hamstring/quadriceps strength ratio is <15% compared to the contralateral side (optional isokinetic strength testing of knee flexors and extensors at 60°, 180° and 300° per second and an endurance test at 180° per second).
- Hop tests > 85% compared to the contralateral side.
- Exercises of previous week are carried out properly, and the patient tolerates sport-specific activities and agility training with maximal duration and speed.
- Administer the IKDC questionnaire again.

Appendix 2: overview of tests

1) Measurement pain with the VAS-score: Measurement:

O Presurgery
O Day 2
O End week 8
O End week 15
O End week 22

The patient indicates the pain level with a vertical line on a 100-mm horizontal line. Left means no pain, right means intolerable, hardly bearable pain. The result in mm will be blinded for the patient.

NO PAIN UNBEARABLE PAIN

VAS-score: …… mm

2) Measurement of knee swelling with measuring tape: Measurement:

O Presurgery
O Day 2
O End week 8
O End week 15
O End week 22

The patient sits with supported back and extended but relaxed knee. The lower side of a non-stretchable measurement tape will be applied 1 cm above the top of the patella, to measure circumference of the knee in mm.

Circumference knee: …… mm

3) ROM measurement with a goniometer: Measurement:

O Presurgery
O Day 2
O End week 8
O End week 15
O End week 22

The patient is in supine position and slides the heel of the involved leg toward the buttock. The maximum amount of knee flexion (in degrees) will be measured with a goniometer (placement of goniometer axis at the lateral joint line). The patient extends the involved leg, after which the maximum amount of extension is measured the same way.

Flexion **: ……** °
Extension **: ……** °

4) IKDC 2000 questionnaire: Measurement:

O Presurgery
O End week 8
O End week 15
O End week 22

Name: ____

Date: ______/_______/______ Date of injury: ______/________/_______
Day Month Year Day Month Year

Symptoms:
Grade symptoms at the highest activity level at which you think you could function without significant symptoms, even if you are not actually performing activities at this level.

1. What is the highest level of activity that you can perform without significant knee pain?
   □ Very strenuous activities like jumping or pivoting as in basketball or soccer
   □ Strenuous activities like heavy physical work, skiing or tennis
   □ Moderate activities like moderate physical work, running or jogging
   □ Light activities like walking, housework or yard work
   □ Unable to perform any of the above activities due to knee pain

2. During the past 4 weeks, or since your injury, how often have you had pain?

	0	1	2	3	4	5	6	7	8	9	10	
Never	□	□	□	□	□	□	□	□	□	□	□	Constant

3. If you have pain, how severe is it?

	0	1	2	3	4	5	6	7	8	9	10	
No pain	□	□	□	□	□	□	□	□	□	□	□	Worst pain imaginable

4. During the past 4 weeks, or since your injury, how stiff or swollen was your knee?
   □ Not at all
   □ Mildly
   □ Moderately
   □ Very
   □ Extremely

5. What is the highest level of activity you can perform without significant swelling in your knee?
   □ Very strenuous activities like jumping or pivoting as in basketball or soccer
   □ Strenuous activities like heavy physical work, skiing or tennis
   □ Moderate activities like moderate physical work, running or jogging
   □ Light activities like walking, housework, or yard work
   □ Unable to perform any of the above activities due to knee swelling

6. During the past 4 weeks, or since your injury, did your knee lock or catch?
   □ Yes □ No

7. What is the highest level of activity you can perform without significant giving way in your knee?
   □ Very strenuous activities like jumping or pivoting as in basketball or soccer
   □ Strenuous activities like heavy physical work, skiing or tennis
   □ Moderate activities like moderate physical work, running or jogging
   □ Light activities like walking, housework or yard work
   □ Unable to perform any of the above activities due to giving way of the knee

Sports activities:
8. What is the highest level of activity you can participate in on a regular basis?
   □ Very strenuous activities like jumping or pivoting as in basketball or soccer
   □ Strenuous activities like heavy physical work, skiing or tennis
   □ Moderate activities like moderate physical work, running or jogging
   □ Light activities like walking, housework or yard work
   □ Unable to perform any of the above activities due to knee

9. How does your knee affect your ability to:

		Not difficult at all	Minimally difficult	Moderately difficult	Extremely difficult	Unable to do
a.	Go up stairs	□	□	□	□	□
b.	Go down stairs	□	□	□	□	□
c.	Kneel on the front of your knee	□	□	□	□	□
d.	Squat	□	□	□	□	□
e.	Sit with your knee bent	□	□	□	□	□
f.	Rise from a chair	□	□	□	□	□
g.	Run straight ahead	□	□	□	□	□
h.	Jump and land on your involved leg	□	□	□	□	□
i.	Stop and start quickly	□	□	□	□	□

Function:
10. How would you rate the function of your knee on a scale of 0 to 10 with 10 being normal, excellent function and 0 being the inability to perform any of your usual daily activities which may include sports?

Function prior to your knee injury:

	0	1	2	3	4	5	6	7	8	9	10	
Cannot perform daily activities	□	□	□	□	□	□	□	□	□	□	□	No limitation in daily activities

Current function of your knee:

	0	1	2	3	4	5	6	7	8	9	10	
Cannot perform daily activities	□	□	□	□	□	□	□	□	□	□	□	No limitation in daily activities

IKDC score: ……

The IKDC Subjective Knee Evaluation Form is scored by summing the scores for the individual items and then transforming the score into a scale that ranges from 0-to-100. Note: The response to item 10 "Function Prior to Knee Injury" is not included in the overall score. The steps to score the IKDC Subjective Knee Evaluation Form are as follows:

1. Assign a score to the individual's response for each item, such that lowest score represents the lowest level of function or highest level of symptoms (i.e., question 1: 1 to 5 points, question 2: 1 to 11 points, question 6: 1 to 2 points).

2. Calculate the raw score by summing the responses to all items with the exception of the response to item 10 "Function Prior to Your Knee Injury".

3. Transform the raw score into a 0-to-100 scale as follows:

$$\text{IKDC Score} = \frac{\text{Raw Score - Lowest Possible Score}}{\text{Range of Scores}} \times 100$$

Where the lowest possible score is 18 and the range of possible scores is 87 (105–18). Thus, if the sum of scores for the 18 items is 60, the IKDC Score would be calculated as follows:

$$\text{IKDC Score} = \frac{60-18}{87} \times 100 = 48.3$$

The transformed score is interpreted as a measure of function such that higher scores represent higher levels of function and lower levels of symptoms. A score of 100 is interpreted to mean no limitation with activities of daily living or sports activities and the absence of symptoms. The IKDC Subjective Knee Score can still be calculated if there are missing data, as long as there are responses to at least 90% of the items (i.e., responses have been provided for at least 16 items). To calculate the raw IKDC score when there are missing data, substitute the average score of the items that have been answered for the missing item score(s). Once the raw IKDC score has been calculated, it is transformed into the IKDC Subjective Knee Score as described above.

5) Hop tests (functional tests): Measurement: O Presurgery
O End week 15
O End week 22

One-legged time hop test:
Instruction: hop 6 meters as quickly as possible (with big powerful hops) without arm swing (keep the arms on your back). First with the involved leg, followed by the uninvolved leg. This test will be performed twice.
Calculation: the average of the two times of the uninvolved leg will be divided by the average of the two times of the involved leg and multiplied by 100.

One-legged time hop test: ……%

Single hop for distance:
Instruction: 1 hop as far as possible, without arm swing (keep the arms on your back) and stand still for at least two seconds. First with the involved leg, followed by the uninvolved leg. This test will be performed twice.
Calculation: the average of the two distances of the involved leg will be divided by the average of the two distances of the uninvolved leg and multiplied by 100.

Single hop for distance: ……%

Triple hop for distance:
Instruction: 3 hops as far as possible, without arm swing (keep the arms on your back) and stand still for at least two seconds. First with the involved leg, followed by the uninvolved leg. This test will be performed twice.
Calculation: the sum of the two distances of the involved leg will be divided by the sum of the two distances of the uninvolved leg and multiplied by 100.

Triple hop for distance: ……%

6) Optional isokinetic tests (quadriceps and hamstring): Measurement: O Presurgery
O End week 15
O End week 22

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